Structure and Function of the Cardiovascular System

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Perfusion → Heart wall → Blood flow through the heart → Valves → Vessels → Coronary arteries → Conduction system and action potential → Cardiac cycle → Cardiac output → Blood pressure regulation → Nursing assessment

This lecture covers normal cardiovascular physiology and how healthy mechanics show up in a clinical assessment. The cardiovascular system builds on fluid and electrolyte balance, and it connects to the endocrine and renal systems as risk factors, causes, or effects on perfusion. Mastering normal function gives you the foundation for understanding what happens when the system breaks down.

Perfusion

Perfusion is the delivery of oxygenated blood and nutrients to the tissue beds to maintain cellular function. Everything assessed in the cardiovascular system centers on it.

Flow = Pressure gradient ÷ Resistance

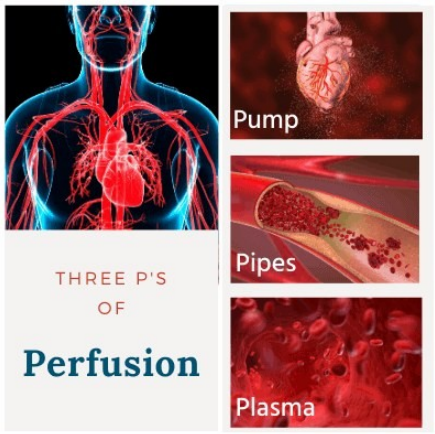
The pressure gradient is generated by the heart's contraction. Flow is inversely related to vascular resistance (the pressure created by the blood vessels). For blood to flow out of the heart, the heart must generate a higher pressure than the vasculature, moving blood from high pressure to low pressure.

Bedside Evidence of Normal Perfusion

- Warm extremities
- Brisk capillary refill (3 seconds or less)
- Clear mentation and normal level of consciousness
- Strong, equal peripheral pulses (2+ bilaterally)
- Adequate urine output, which shows the kidneys are being perfused

Exam note: The renal exam is on the same test: the goal urine output is at least 0.5 mL/kg/hr.

Keep asking why you see these findings: what physiological cause explains each sign of adequate perfusion?

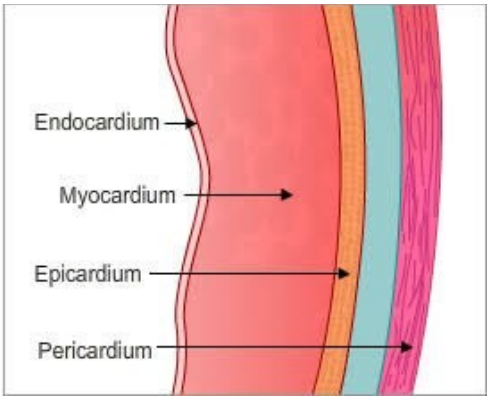


The three P's of perfusion: pump, pipes, plasma (lecture slide)

The Heart Wall

The heart wall has four layers, from the outside in:

- **Pericardium:** a protective double-walled sac holding about 15–50 mL of lubricating fluid, letting the heart contract without friction (like the pleural cavity around the lungs).
- **Epicardium:** the outer visceral layer directly covering the heart.
- **Myocardium:** the muscular layer responsible for mechanical contraction. When we discuss damaged or thickened heart muscle, this is the layer involved.
- **Endocardium:** the smooth inner lining of the heart chambers, extending over the valves.



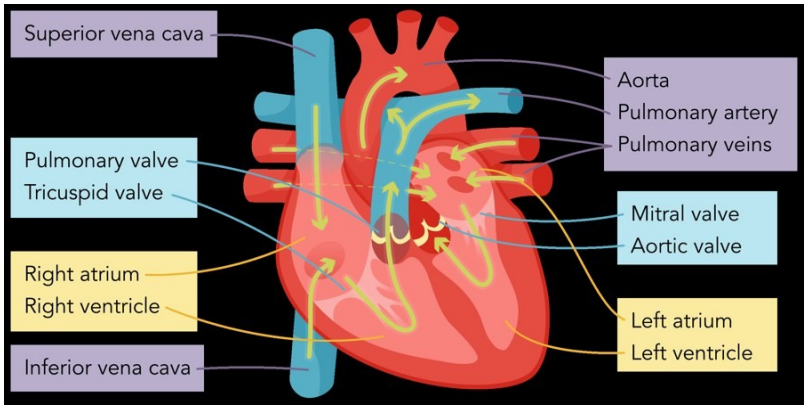
Layers of the heart wall (lecture slide)

Blood Flow Through the Heart

This is review from anatomy and physiology; nothing has changed. The heart has two circuits: pulmonary (to the lungs) and systemic (to the body).

	Pulmonary circulation	Systemic circulation
Pressure / resistance	Low pressure, low resistance	High pressure, high resistance
Pathway	Vena cava (superior and inferior) → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary arteries → lungs for gas exchange	Pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta → body
Special note	Pulmonary arteries are the only arteries that carry deoxygenated blood	Pulmonary veins are the only veins that carry oxygenated blood

The left ventricle pumps against higher systemic resistance than the right, so its myocardium is naturally about 3 times thicker than the right ventricle's.

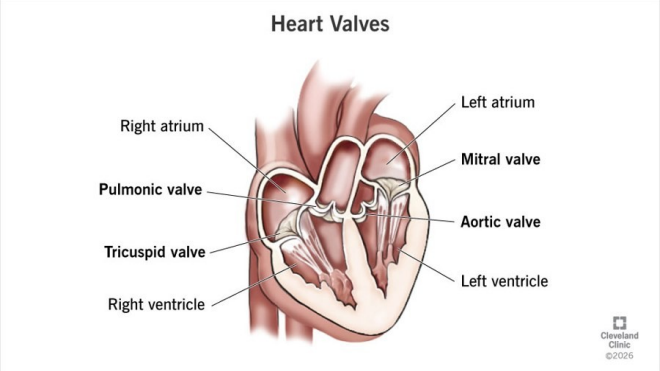


Heart structures and blood flow (lecture slide)

Heart Valves

Four valves ensure one-way blood flow.

	Atrioventricular (AV) valves	Semilunar valves
Valves	Tricuspid (right) and mitral (left)	Aortic and pulmonic
Location	Between atria and ventricles	At the exits of the ventricles
Mechanism	Tethered by chordae tendineae and papillary muscles. When ventricular pressure rises, papillary muscles pull the chordae taut so leaflets cannot bulge back into the atria. The valves stay closed during ventricular contraction.	Open purely in response to forward pressure during ventricular contraction. They snap shut when ventricular pressure falls below the vascular pressure.



Heart valves (lecture slide)

The Vascular System

Arteries and veins both have three tissue layers:

- **Tunica intima:** smooth endothelial lining (innermost).
- **Tunica media:** smooth muscle and elastic tissue (middle). Arteries have a much thicker media than veins to handle their high hydrostatic pressures.
- **Tunica externa:** supportive outer connective tissue.

Arteries

Arteries carry blood away from the heart, and it is oxygenated blood except in the pulmonary arteries. Types:

- **Elastic arteries (e.g., the aorta):** expand and recoil with each beat.
- **Muscular arteries and arterioles:** arterioles contain more smooth muscle, so they adapt readily to pressure changes by constricting and dilating.

Veins

Veins return deoxygenated blood to the heart, except the pulmonary veins, which carry oxygenated blood from the lungs to the left atrium.

- **Superficial veins:** near the skin surface; visible.
- **Deep veins:** deeper in the body; not visible.

Venous pressure is lower than arterial pressure, so veins rely on added mechanisms: the muscle pump, which "milks" blood back toward the heart, and the respiratory pump. Some veins, particularly in the lower extremities, have one-way valves that keep blood moving toward the heart and prevent pooling and backflow.

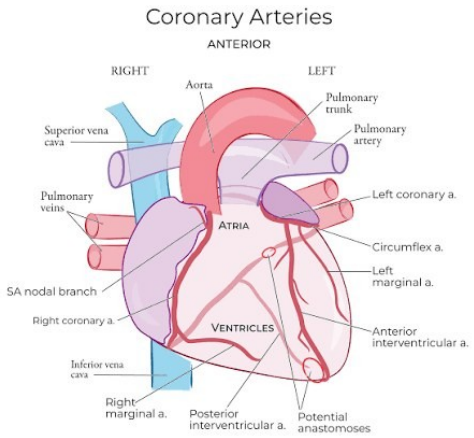
Coronary Arteries

The heart has its own blood supply through three main coronary arteries:

- **Left anterior descending (LAD) and circumflex arteries:** supply the rest of the heart (the left side).
- **Right coronary artery (RCA):** supplies the right side of the heart and the inferior wall of the left ventricle.

Key principle: The heart perfuses its own tissue during DIASTOLE, not systole. In systole the contracting muscle compresses the coronary arteries and blocks flow. In diastole, blood flows freely into them. At a normal resting rate (60–100 bpm), diastole is long enough to keep the myocardium well oxygenated.

Lack of blood flow to the heart can cause major problems, and this will matter as we study coronary disorders.



Coronary arteries (lecture slide)

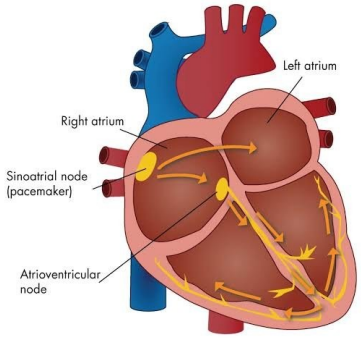
Conduction System and Action Potential

Exam note: EKGs will NOT be on your test. This is an introduction so that next semester's EKG content feels familiar.

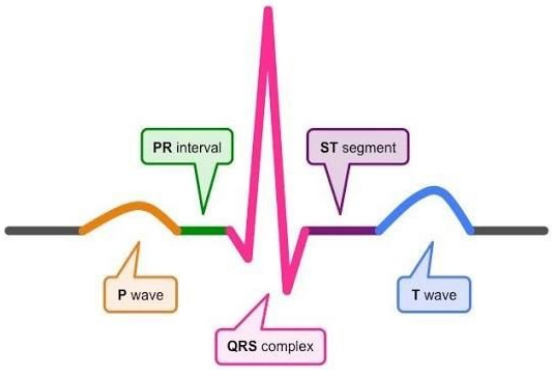
Mechanical movement in the heart is always preceded by electrical depolarization. The conduction sequence follows a precise pathway:

Structure	Intrinsic rate	Role and EKG correlate
SA node (top of right atrium)	60–100 bpm	Dominant pacemaker; sets the normal heart rate. Produces a clean P wave before every QRS (atrial contraction).
AV node	40–60 bpm	Introduces a brief delay, giving the atria time to finish contracting and fill the ventricles. Corresponds to the flat (isometric) PR segment.
Bundle of His, bundle branches		At the end of the PR interval, the signal leaves the AV node and travels down the ventricular septum.
Purkinje fibers	20–40 bpm	Wrap around the ventricles and spread the impulse rapidly through the ventricular myocardium, producing the tall, narrow QRS complex (a quick, strong impulse for a strong contraction).

- **Backup rate:** the Purkinje rate (20–40 bpm) is a backup if the SA and AV nodes both fail, and is not compatible with life on its own. Athletes with a low resting rate still have it driven by the SA node, not the Purkinje fibers.
- **Repolarization:** ventricular repolarization (rest) follows the QRS. Atrial repolarization is hidden by the large QRS complex and is not seen on the EKG.



Cardiac conduction pathway (lecture slide)



Normal EKG waveform (lecture slide)

Cardiac Action Potential

At the cellular level, the action potential drives these electrical impulses, in five phases:

Phase	What happens
0: Rapid depolarization	Fast sodium channels open; surge of sodium into the cell.
1: Initial repolarization	Sodium channels close and potassium begins leaving the cell.
2: Plateau	Slow calcium channels open, so calcium flows in while potassium continues to exit. The calcium influx triggers muscle contraction.
3: Repolarization	Calcium channels close and potassium rapidly exits, restoring the negative membrane potential. Seen as the T wave.
4: Resting	The sodium-potassium ATPase pump restores baseline ion concentrations and the cycle restarts.

Because potassium and calcium shape the action potential, they affect cardiac function and conduction, which is why electrolyte changes appear on the EKG.

The Cardiac Cycle

Electrical impulses connect to the mechanical cardiac cycle:

1. **Atrial systole / late diastole:** the ventricles fill passively to about 70–80% capacity. The atria then contract and deliver the remaining 20–30%, called the atrial kick.
2. **Isovolumetric contraction:** ventricular pressure exceeds atrial pressure and snaps the AV valves shut. This closure creates the S1 sound ("lub"). For a fraction of a second all four valves are closed.
3. **Ejection:** as pressure rises rapidly with volume constant, ventricular pressure exceeds aortic and pulmonary artery pressure, pushing the semilunar valves open to eject blood.
4. **Isovolumetric relaxation:** the ventricles relax, pressure falls below vascular pressure, and blood begins to flow backward, snapping the semilunar valves shut. This creates the S2 sound ("dub").

Remember: S1 and S2 are the sounds of valves closing. This is what you hear with a stethoscope, and why murmurs (extra sounds) point to valve problems. Listen for crisp S1 and S2 to confirm normal valve closure.

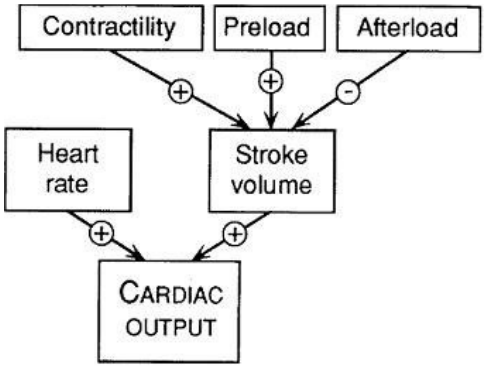
Cardiac Output

Cardiac output (CO) is the amount of blood pumped from the heart to the systemic circulation each minute.

$$CO = \text{Heart rate} \times \text{Stroke volume}$$

Normal resting CO is about 4–8 L/min. Stroke volume (SV) is the volume ejected per beat, about 70 mL, and has three determinants:

- **Preload:** the volume of blood in the ventricles at the end of diastole. By the Frank-Starling law, greater stretch during filling produces a stronger contraction, up to a physiological limit. Think of a rubber band or balloon: pulled back further it snaps harder, but stretched too far it loses elasticity and the contraction weakens.
- **Afterload:** the resistance the ventricles must overcome to eject blood. For the left ventricle it is mainly systemic vascular tone; for the right ventricle, pulmonary vascular resistance. The left side faces a higher pressure than the right.
- **Contractility:** the intrinsic force of the muscle contraction, influenced by sympathetic stimulation and available intracellular calcium. We want a quick, strong contraction to get blood out to the body.



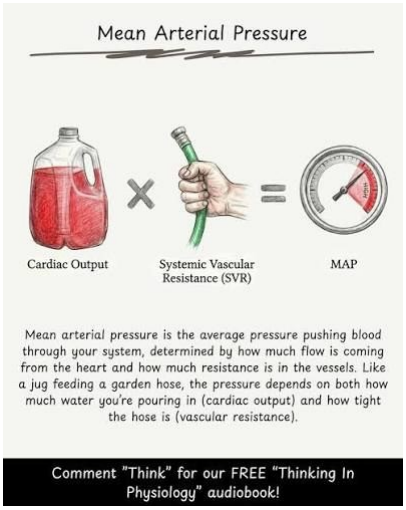
Determinants of cardiac output (lecture slide)

Vascular Physiology and Autonomic Control

MAP = Cardiac output × Systemic vascular resistance

Mean arterial pressure (MAP) reflects systemic vascular pressure. A healthy adult MAP is about 70–105 mmHg, and 65 mmHg or higher is acceptable to ensure organ perfusion. It can also be calculated from blood pressure (MAP = DBP + ⅓(SBP − DBP)), and automated BP machines report it along with systolic and diastolic pressures.

MAP is often used to set blood pressure goals for patients with low pressure, such as those with systemic infection or hypovolemia. Normal MAP depends largely on patient size: the shorter the patient, the lower the MAP, which matters in pediatrics versus adults. The adult goal is still 70–105.



Mean arterial pressure = CO × SVR (lecture slide)

Feedback Loops

The body maintains this baseline through short-term and long-term loops.

- **Baroreceptors (short-term autonomic reflex):** stretch receptors in the carotid sinuses and aortic arch continuously monitor arterial wall stretch.
  - **If BP drops:** less stretch signals the medulla to increase sympathetic activity: higher heart rate, stronger contractility, and constricted arterioles.
  - **If BP rises:** more stretch increases parasympathetic tone, slowing the heart rate and lowering vascular resistance.

- **RAS (long-term hormonal):** the kidneys sense reduced perfusion and release renin, which starts a cascade producing angiotensin II (a systemic vasoconstrictor) and aldosterone (promotes renal sodium and water retention to preserve blood volume). Together they raise pressure through vasoconstriction and fluid retention, as long as vasoconstriction and volume are adequate.
- **Natriuretic peptides (BNP, ANP):** released by atrial and ventricular myocardial cells when wall stretch rises from increased fluid volume. They signal the kidneys to excrete sodium and water (water follows salt), keeping volume normal and countering volume overload. We focus on BNP in the cardiac system.

Nursing Assessment: What Is Normal?

Keep these normals in mind as you study disorders of each system. A value outside the normal range may still be a patient's baseline, so assess the rest of the perfusion picture and ask: is this new?

Assessment	Normal finding
Heart rate and rhythm	60–100 bpm with a regular, consistent rhythm on the monitor (not beats clustered close together then spread apart)
Heart sounds	Crisp S1 and S2 with no extra sounds (no whooshing or clicking)
Perfusion indicators	Warm skin; capillary refill 3 seconds or less; strong, equal bilateral 2+ pulses; steady urine output (0.5 mL/kg/hr)
Autonomic balance	Stable BP with MAP ≥ 65 mmHg (about 65–105), adapting appropriately to position changes and activity

Notes on Transcript Cleanup

Corrected mis-transcriptions: "packing"/"congi" fibers are Purkinje fibers; "bundle of his" is bundle of His; "chordae tendineais" is chordae tendineae; "arteriolars" is arterioles; "artery veins" is pulmonary veins; "intercellular calcium" is intracellular; "basic natriuretic peptide" is B-type (brain) natriuretic peptide. Where the speaker misspoke (for example, blood flow going "to the heart" instead of the lungs, or a comparison of left side to left side), I used the sense of the surrounding explanation. The action potential phase numbers (0–4) follow the standard convention and were not stated in the lecture.